

Insects

PHYLUM	Arthropoda
SUBPHYLUM	Mandibulata
SUPERCLASS	Hexapoda
CLASS	Insecta
ORDERS	29
FAMILIES	About 1,000
SPECIES	About 1.1 million

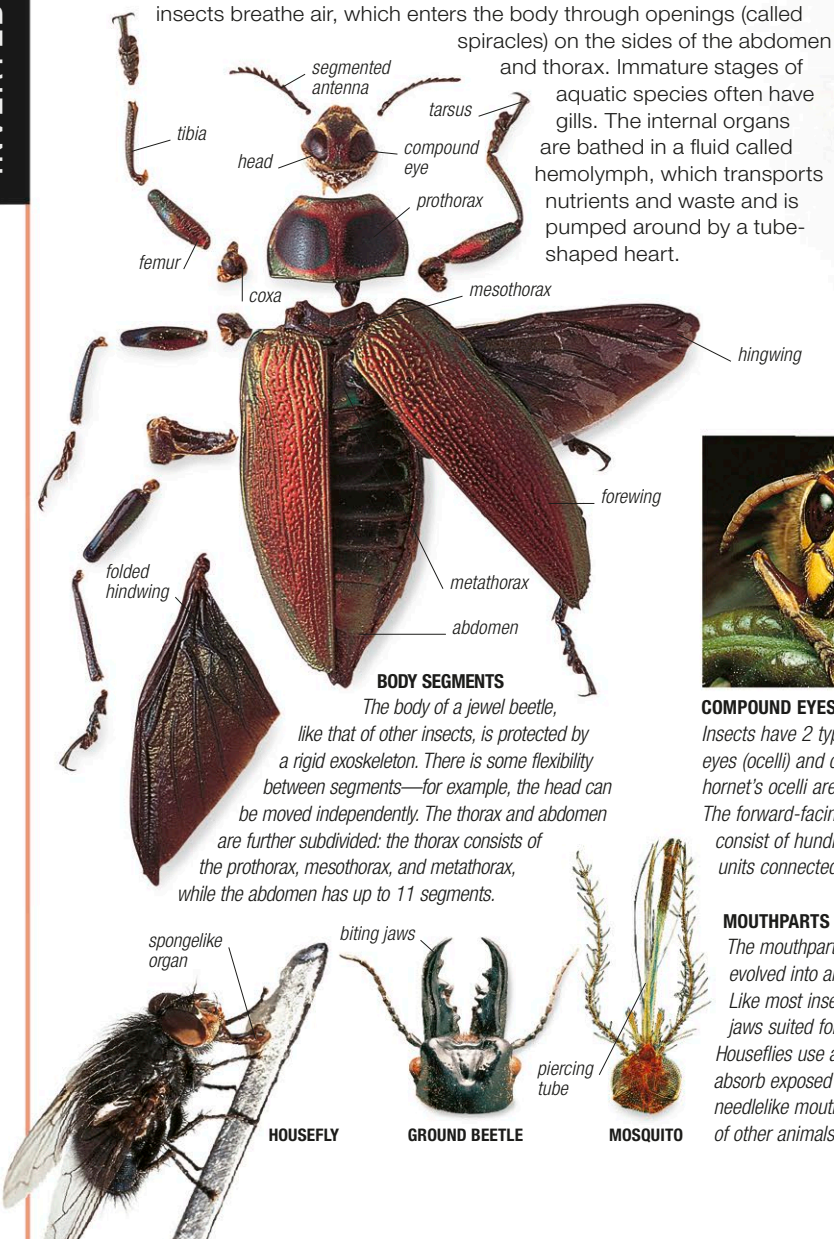
Insects are the most successful animals on earth. They account for more species than any other class: over a million have so far been identified, but it is thought that the true number may be between 5 and 10 million. Insects belong to a group of arthropods called hexapods, all of which have 3 pairs of legs. Many insects also have wings, which makes them the only arthropods capable of powered flight.

Combined with their small size and ability to survive in dry environments, flight has enabled insects to colonize a vast range of habitats. Most live on land or in the air, but they are also numerous in freshwater. Without insects, many other forms of life would not exist: the majority of flowering plants are largely dependent on them for pollination, and they form the main component of the diet of many animals. During their lifetime, many insects undergo complete metamorphosis, passing through several physical stages before reaching adulthood.

Anatomy

The body of an insect is divided into 3 segments—the head, thorax, and abdomen—each of which performs different functions. The head supports the mouthparts and much of the sensory apparatus, including the compound eyes and antennae. The thorax, which bears the legs and in many species the wings, is important in locomotion, while the abdomen contains the organs for digestion, excretion, and reproduction. All adult

insects breathe air, which enters the body through openings (called spiracles) on the sides of the abdomen and thorax. Immature stages of aquatic species often have gills. The internal organs are bathed in a fluid called hemolymph, which transports nutrients and waste and is pumped around by a tube-shaped heart.



BODY SEGMENTS

The body of a jewel beetle, like that of other insects, is protected by a rigid exoskeleton. There is some flexibility between segments—for example, the head can be moved independently. The thorax and abdomen are further subdivided: the thorax consists of the prothorax, mesothorax, and metathorax, while the abdomen has up to 11 segments.

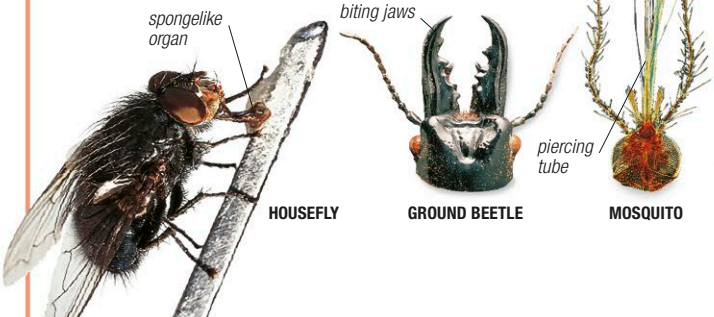


COMPOUND EYES

Insects have 2 types of eyes: simple eyes (ocelli) and compound eyes. This hornet's ocelli are on top of its head. The forward-facing compound eyes consist of hundreds of light-sensitive units connected by nerves to the brain.

MOUTHPARTS

The mouthparts of insects have evolved into an amazing array of forms. Like most insects, ground beetles have jaws suited for cutting and chewing. Houseflies use a sponglike structure to absorb exposed liquid. Mosquitoes use needlelike mouthparts to pierce the skin of other animals.



NON-INSECT HEXAPODS

In addition to insects, hexapods include 3 smaller classes: springtails, proturans, and diplurans. They are collectively known as non-insect hexapods, and a selection are described in this section. All non-insect hexapods are without wings, and many also lack eyes and antennae. Unlike insects, their mouthparts are enclosed in a pouch located on the underside of their head.



SPRINGTAILS

Springtails are small hexapods that occur in great numbers in soil and leaf litter all over the world, including the Arctic and Antarctic.

Population

There are several reasons why insects have become so successful. Being small, they are able to occupy microhabitats that are inaccessible to other animals. Given the right conditions, they are also able to breed very rapidly, enabling them to respond quickly to increases in the availability of food. For example, a single pair of bruchid beetles have the theoretical potential to produce enough offspring to occupy the entire volume of the earth within 432 days. In practice, this doesn't happen because there are limits to the food supply and because of competition between individuals of the same species and between different species.



SPECIES NUMBERS

High reproduction rates have enabled insects to evolve so rapidly that they make up over half of all animal species alive today. Bugs (such as these firebugs) alone account for 88,000 species. Insects are found on land, in freshwater, and even on the ocean surface.